

P-SV Wave Reflection and Transmission (R/T) Coefficients

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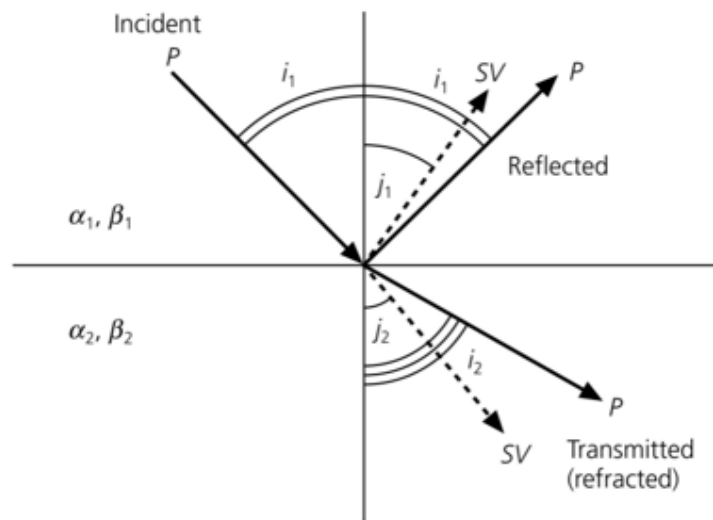


Figure 1: P-SV R/T Coefficients

This note covers **P-SV wave reflection and transmission (R/T) coefficients**, from the full Zoeppritz equations to their linearized form (Shuey's equation) and the special case of the free surface. The P-SV problem is significantly more complex than the SH-case because P-waves and SV-waves couple at an interface. When a P-wave is incident on an interface between two elastic media, it generates four waves (1):

- **Reflected P-wave** (travels back in medium 1)
- **Reflected SV-wave** (mode conversion in medium 1)
- **Transmitted P-wave** (refracted into medium 2)
- **Transmitted SV-wave** (mode conversion in medium 2)

This is fundamentally different from SH waves, where no mode conversion occurs because SH motion is perpendicular to the plane of incidence. This leads to a system of four equations instead of two.

1. Concept Overview of P-Sv Wave Propagation

Problem Setup

We consider an interface between two elastic solids:

- **Medium 1:** P-wave velocity α_1 , S-wave velocity β_1 , density ρ_1 , Lamé parameters λ_1, μ_1
- **Medium 2:** P-wave velocity α_2 , S-wave velocity β_2 , density ρ_2 , Lamé parameters λ_2, μ_2

Snell's Law

For an incident plane wave (P or SV), Snell's Law governs of all generated waves:

$$\frac{\sin i}{\alpha_1} = \frac{\sin i'}{\alpha_1} = \frac{\sin j}{\beta_1} = \frac{\sin i_2}{\alpha_2} = \frac{\sin j_2}{\beta_2} = p$$

where p is the ray parameter, constant for all waves.

Potential Functions

To handle the coupled P-SV system, we use scalar potential ϕ for P-waves and vector potential ψ for SV-waves, where the displacement $\mathbf{u}$ is:

$$\mathbf{u} = \nabla\phi + \nabla \times \psi$$

P-waves (dilatational):

$$\mathbf{u}_P = \nabla\phi \tag{1}$$

SV-waves (shear, vertical polarization):

$$\mathbf{u}_{SV} = \nabla \times (0, \psi, 0) \tag{2}$$

For 2D P-SV case in x-z plane, the vector potential has only a y-component: $\psi = (0, \psi, 0)$.

Boundary Conditions

At interface $z = 0$, we require continuity of:

1. **Displacement:** u_x and u_z

$$\begin{pmatrix} u_x \\ u_z \end{pmatrix} = \begin{pmatrix} \partial_x & -\partial_z \\ \partial_z & \partial_x \end{pmatrix} \begin{pmatrix} \phi \\ \psi \end{pmatrix} \tag{3}$$

This leads to the displacement components:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z}, \quad u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x}$$

2. **Stress:** σ_{zz} (normal stress) and σ_{xz} (shear stress)

Stress expressions in terms of potentials:

Remember:

$$\sigma_{ij} = \lambda \delta_{ij} \epsilon_{kk} + 2\mu \epsilon_{ij}$$

$$\epsilon_{kk} = \nabla \cdot u = \epsilon_{11} + \epsilon_{22} + \epsilon_{33}$$

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right)$$

$$\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2} \right)$$

The Zoeppritz Equations:

Applying these four boundary conditions to the complete wavefield (incident P, reflected P, reflected SV, transmitted P, transmitted SV) results in a 4x4 system of linear equations. This full system is the Zoeppritz equations. Solving them gives the exact R/T coefficients for a planar interface between two solids. They are the "ground truth" but are algebraically complex and non-intuitive.

2. Wave Equations in Each Medium

Medium 1:

$$\phi_1 = A_1 e^{i(\omega t - k_x x + k_{z\alpha} z)} + A' e^{i(\omega t - k_x x - k_{z\alpha} z)} \quad (4)$$

$$\psi_1 = B_1 e^{i(\omega t - k_x x - k_{z\beta} z)} \quad (5)$$

Medium 2:

$$\phi_2 = A_2 e^{i(\omega t - k_x x + k_{z\alpha} z)} \quad (6)$$

$$\psi_2 = B_2 e^{i(\omega t - k_x x + k_{z\beta} z)} \quad (7)$$

Sign conventions:

- Positive k_z indicates downward propagation
- Negative k_z indicates upward propagation

Wavenumber Relations

From Snell's law, k_x is continuous across the interface:

$$k_{z\alpha} = \sqrt{\frac{\omega^2}{\alpha^2} - k_x^2}, \quad k_{z\beta} = \sqrt{\frac{\omega^2}{\beta^2} - k_x^2} \quad (8)$$

3. Displacement Fields

Computing Displacements at Interface from Potentials

To handle the coupled P-SV system, we use scalar potential ϕ for P-waves and vector potential ψ for SV-waves, where the displacement $\mathbf{u}$ is:

$$\mathbf{u} = \nabla \phi + \nabla \times \psi$$

P-waves (dilatational):

P-waves are **irrotational** (no rotation) and can be described by a scalar potential ϕ such that:

$$\mathbf{u}_P = \nabla \phi \quad (9)$$

In component form:

$$\boxed{u_x = \frac{\partial \phi}{\partial x}, \quad u_z = \frac{\partial \phi}{\partial z}} \quad (10)$$

No u_y component because we are considering 2D.

Take Note:

Why this works

Wave equation for P-waves

$$\nabla^2 \phi = \frac{1}{\alpha^2} \frac{\partial^2 \phi}{\partial t^2}$$

Checking irrotationality

$$\nabla \times \mathbf{u}_P = \nabla \times (\nabla \phi) = 0$$

This is always true because the curl of a gradient is identically zero.

Dilatation (volume change)

$$\nabla \cdot \mathbf{u}_P = \nabla \cdot (\nabla \phi) = \nabla^2 \phi \neq 0$$

SV-waves (shear, vertical polarization):

SV-waves are **solenoidal** (divergence-free) and can be described by a vector potential. For 2D motion in the x-z plane, we use:

$$\mathbf{u}_{SV} = \nabla \times (0, \psi, 0) = \nabla \times (\psi \hat{y}) \quad (11)$$

For 2D P-SV case in x-z plane, the vector potential has only a y-component: $\psi = (0, \psi, 0)$.

Detailed derivation

The curl in Cartesian coordinates is:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (12)$$

For $\mathbf{A} = (0, \psi, 0)$:

$$\nabla \times (0, \psi, 0) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \psi & 0 \end{vmatrix} \quad (13)$$

Computing each component:

x-component:

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} = \frac{\partial 0}{\partial y} - \frac{\partial \psi}{\partial z} = -\frac{\partial \psi}{\partial z} \quad (14)$$

For 2D problems where nothing depends on y ($\partial/\partial y = 0$):

Standard curl calculation:

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} = -\frac{\partial \psi}{\partial z} \quad (15)$$

$$(\nabla \times \mathbf{A})_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} = 0 \quad (16)$$

$$(\nabla \times \mathbf{A})_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} = \frac{\partial \psi}{\partial x} \quad (17)$$

$$u_x = -\frac{\partial \psi}{\partial z}, \quad u_z = \frac{\partial \psi}{\partial x} \quad (18)$$

So:

$$\nabla \times \boldsymbol{\psi} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \psi & 0 \end{vmatrix} = -\hat{x} \frac{\partial \psi}{\partial z} + \hat{z} \frac{\partial \psi}{\partial x} \quad (19)$$

Therefore:

$$\boxed{u_x = -\frac{\partial \psi}{\partial z}, \quad u_z = +\frac{\partial \psi}{\partial x}} \quad (20)$$

Why this works

Wave equation for SV-waves

$$\nabla^2 \psi = \frac{1}{\beta^2} \frac{\partial^2 \psi}{\partial t^2} \quad (21)$$

Checking solenoidality (divergence-free)

$$\nabla \cdot \mathbf{u}_{SV} = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} = \frac{\partial^2 \psi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z \partial x} = 0 \quad (22)$$

SV-waves involve no volume change.

Rotation (vorticity)

$$\nabla \times \mathbf{u}_{SV} = \nabla \times (\nabla \times (0, \psi, 0)) \neq 0 \quad (23)$$

SV-waves involve rotation/shear.

So, from equation 10 and 20:

Horizontal displacement:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \quad (24)$$

Vertical displacement:

$$u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \quad (25)$$

In matrix form, the continuity of displacement at the interface becomes:

$$\boxed{\begin{pmatrix} u_x \\ u_z \end{pmatrix} = \begin{pmatrix} \partial_x & -\partial_z \\ \partial_z & \partial_x \end{pmatrix} \begin{pmatrix} \phi \\ \psi \end{pmatrix}} \quad (26)$$

4. Computing Stress Components in Terms of Potentials

General Stress-Strain Relations

For an isotropic elastic medium, the stress tensor is:

$$\sigma_{ij} = \lambda \delta_{ij} \epsilon_{kk} + 2\mu \epsilon_{ij} \quad (27)$$

where:

- λ, μ are Lamé parameters
- δ_{ij} is the Kronecker delta
- $\epsilon_{kk} = \epsilon_{11} + \epsilon_{22} + \epsilon_{33}$ is the dilatation (volumetric strain)
- $\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$ is the strain tensor

Step 1: Express Dilatation in Terms of Displacement

The dilatation is:

$$\epsilon_{kk} = \nabla \cdot \mathbf{u} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \quad (28)$$

For 2D problems in the x-z plane (no y-dependence):

$$\epsilon_{kk} = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} \quad (29)$$

Step 2: Express Displacements in Terms of Potentials

Recall:

- P-wave: $\mathbf{u}_P = \nabla \phi$, so $u_x = \frac{\partial \phi}{\partial x}$, $u_z = \frac{\partial \phi}{\partial z}$
- SV-wave: $\mathbf{u}_{SV} = \nabla \times (0, \psi, 0)$, so $u_x = -\frac{\partial \psi}{\partial z}$, $u_z = \frac{\partial \psi}{\partial x}$

Total displacement:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \quad (30)$$

$$u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \quad (31)$$

Step 3: Calculate Dilatation

$$\epsilon_{kk} = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} \quad (32)$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \right) \quad (33)$$

$$= \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \psi}{\partial x \partial z} + \frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial z \partial x} \quad (34)$$

$$= \frac{\partial^2 \phi}{\partial x^2} - \cancel{\frac{\partial^2 \psi}{\partial x \partial z}} + \frac{\partial^2 \phi}{\partial z^2} + \cancel{\frac{\partial^2 \psi}{\partial z \partial x}} \quad (35)$$

$$\epsilon_{kk} = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi \quad (36)$$

So:

$$\boxed{\epsilon_{kk} = \nabla \cdot \mathbf{u} = \nabla^2 \phi} \quad (37)$$

This shows that the dilatation depends only on the P-wave potential ϕ , not on the SV-wave potential ψ .

Step 4: Derive σ_{zz}

From the general formula (equation 31):

$$\sigma_{zz} = \lambda \epsilon_{kk} + 2\mu \epsilon_{zz} \quad (38)$$

where:

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} \quad (39)$$

Computing ϵ_{zz}

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} = \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \right) = \frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \quad (40)$$

Substituting into σ_{zz}

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right) \quad (41)$$

$$= \lambda \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} \right) + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (42)$$

$$= \lambda \frac{\partial^2 \phi}{\partial x^2} + \lambda \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (43)$$

$$= \lambda \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} \right) + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (44)$$

Note that $\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2}$, So, this can be rewritten as:

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (45)$$

$$\boxed{\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right)} \quad (46)$$

Step 5: Derive σ_{xz}

The shear stress is:

$$\sigma_{xz} = 2\mu \epsilon_{xz} = 2 \cdot \mu \cdot \frac{1}{2} \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) = \mu \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) \quad (47)$$

Computing Each Term

First term:

$$\frac{\partial u_x}{\partial z} = \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \right) = \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} \quad (48)$$

Second term:

$$\frac{\partial u_z}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \right) = \frac{\partial^2 \phi}{\partial z \partial x} + \frac{\partial^2 \psi}{\partial x^2} \quad (49)$$

Adding the Terms

$$\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} = \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} \quad (50)$$

$$= 2 \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x^2} \quad (51)$$

Final Expression for σ_{xz}

$$\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x^2} \right) \quad (52)$$

$$\boxed{\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2} \right)} \quad (53)$$

The fundamental relationships are:

$$\boxed{\epsilon_{kk} = \nabla \cdot \mathbf{u} = \epsilon_{11} + \epsilon_{22} + \epsilon_{33} = \nabla^2 \phi} \quad (54)$$

$$\boxed{\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right) = \lambda (\partial_{xx} \phi + \partial_{zz} \phi) + 2\mu (\partial_{zz} \phi + \partial_{xz} \psi)} \quad (55)$$

$$\boxed{\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2} \right) = \mu (2\partial_{xz} \phi + \partial_{xx} \psi - \partial_{zz} \psi)} \quad (56)$$

Physical Interpretation

- The dilatation ϵ_{kk} depends only on the P-wave potential ϕ , confirming that only P-waves cause volume changes.
- The normal stress σ_{zz} involves both ϕ and ψ , showing coupling between P and SV waves at boundaries.
- The shear stress σ_{xz} also involves both potentials, demonstrating mode conversion.
- These expressions allow us to write boundary conditions entirely in terms of ϕ and ψ , which is essential for deriving the Zoeppritz equations.

5. Boundary Conditions at $z=0$

At an interface (say $z = 0$), we require:

Four Boundary Conditions:

1. Continuity of horizontal displacement: $u_x^{(1)} = u_x^{(2)}$
2. Continuity of vertical displacement: $u_z^{(1)} = u_z^{(2)}$
3. Continuity of shear stress: $\sigma_{xz}^{(1)} = \sigma_{xz}^{(2)}$
4. Continuity of normal stress: $\sigma_{zz}^{(1)} = \sigma_{zz}^{(2)}$

At a free surface ($z = 0$), we require:

1. $\sigma_{xz} = 0$
2. $\sigma_{zz} = 0$

These conditions, expressed in terms of ϕ and ψ using the formulas derived above, lead to the famous Zoeppritz equations for reflection and transmission coefficients. These four equations also determine the four unknowns: A' , B_1 , A_2 , B_2 .

6. Detailed Derivation of Boundary Conditions

Computing Derivatives

For $\phi_1 = A_1 e^{i(\omega t - k_x x + k_{z\alpha_1} z)} + A' e^{i(\omega t - k_x x - k_{z\alpha_1} z)}$ at $z = 0$:

$$\left. \frac{\partial \phi_1}{\partial x} \right|_{z=0} = -ik_x (A_1 + A') e^{i(\omega t - k_x x)} \quad (57)$$

$$\left. \frac{\partial \phi_1}{\partial z} \right|_{z=0} = ik_{z\alpha_1} (A_1 - A') e^{i(\omega t - k_x x)} \quad (58)$$

For $\psi_1 = B_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)}$ at $z = 0$:

$$\left. \frac{\partial \psi_1}{\partial x} \right|_{z=0} = -ik_x B_1 e^{i(\omega t - k_x x)} \quad (59)$$

$$\left. \frac{\partial \psi_1}{\partial z} \right|_{z=0} = -ik_{z\beta_1} B_1 e^{i(\omega t - k_x x)} \quad (60)$$

Similarly for medium 2 at $z = 0$:

$$\phi_2 = A_2 e^{i(\omega t - k_x x + k_{z\alpha_2} z)} \quad (61)$$

$$\psi_2 = B_2 e^{i(\omega t - k_x x + k_{z\beta_2} z)} \quad (62)$$

$$\left. \frac{\partial \phi_2}{\partial x} \right|_{z=0} = -ik_x A_2 e^{i(\omega t - k_x x)} \quad (63)$$

$$\left. \frac{\partial \phi_2}{\partial z} \right|_{z=0} = ik_{z\alpha_2} A_2 e^{i(\omega t - k_x x)} \quad (64)$$

$$\left. \frac{\partial \psi_2}{\partial x} \right|_{z=0} = -ik_x B_2 e^{i(\omega t - k_x x)} \quad (65)$$

$$\left. \frac{\partial \psi_2}{\partial z} \right|_{z=0} = ik_{z\beta_2} B_2 e^{i(\omega t - k_x x)} \quad (66)$$

Boundary Condition 1: Horizontal Displacement

$$u_x^{(1)} \Big|_{z=0} = u_x^{(2)} \Big|_{z=0} \quad (67)$$

$$\left. \frac{\partial \phi_1}{\partial x} - \frac{\partial \psi_1}{\partial z} \right|_{z=0} = \left. \frac{\partial \phi_2}{\partial x} - \frac{\partial \psi_2}{\partial z} \right|_{z=0} \quad (68)$$

$$-ik_x(A_1 + A') - (-ik_{z\beta_1} B_1) = -ik_x A_2 - ik_{z\beta_2} B_2 \quad (69)$$

Simplifying:

$$k_x(A_1 + A') - k_{z\beta_1} B_1 = k_x A_2 + k_{z\beta_2} B_2 \quad (70)$$

Rearrange:

$$k_x A' - k_{z\beta_1} B_1 - k_x A_2 - k_{z\beta_2} B_2 = -k_x A_1$$

Boundary Condition 2: Vertical Displacement

$$u_z^{(1)} \Big|_{z=0} = u_z^{(2)} \Big|_{z=0} \quad (71)$$

$$\left. \frac{\partial \phi_1}{\partial z} + \frac{\partial \psi_1}{\partial x} \right|_{z=0} = \left. \frac{\partial \phi_2}{\partial z} + \frac{\partial \psi_2}{\partial x} \right|_{z=0} \quad (72)$$

$$ik_{z\alpha_1}(A_1 - A') + (-ik_x B_1) = ik_{z\alpha_2} A_2 + (-ik_x B_2) \quad (73)$$

Simplifying:

$$k_{z\alpha_1}(A_1 - A') - k_x B_1 = k_{z\alpha_2} A_2 - k_x B_2 \quad (74)$$

$$-k_{z\alpha_1} A' - k_x B_1 - k_{z\alpha_2} A_2 + k_x B_2 = -k_{z\alpha_1} A_1$$

Boundary Condition 3: Shear Stress

$$\sigma_{xz}^{(1)} \Big|_{z=0} = \sigma_{xz}^{(2)} \Big|_{z=0} \quad (75)$$

After computing second derivatives:

$$\mu_1[2\partial_{xz}\phi_1 + \partial_{xx}\psi_1 - \partial_{zz}\psi_1]_{z=0} = \mu_2[2\partial_{xz}\phi_2 + \partial_{xx}\psi_2 - \partial_{zz}\psi_2]_{z=0} \quad (76)$$

Computing the second and mixed derivatives using equation 59 - 68:

$$\partial_{xz}\phi_1|_{z=0} = (-ik_x)(ik_{z\alpha_1})(A_1 - A')e^{i(\omega t - k_x x)} = k_x k_{z\alpha_1}(A_1 - A')e^{i(\omega t - k_x x)} \quad (77)$$

$$\partial_{xx}\psi_1|_{z=0} = (-ik_x)^2 B_1 e^{i(\omega t - k_x x)} = -k_x^2 B_1 e^{i(\omega t - k_x x)} \quad (78)$$

$$\partial_{zz}\psi_1|_{z=0} = (-ik_{z\beta_1})^2 B_1 e^{i(\omega t - k_x x)} = -k_{z\beta_1}^2 B_1 e^{i(\omega t - k_x x)} \quad (79)$$

$$\partial_{xz}\phi_2|_{z=0} = (-ik_x)(ik_{z\alpha_2}) A_2 e^{i(\omega t - k_x x)} = k_x k_{z\alpha_2} A_2 e^{i(\omega t - k_x x)} \quad (80)$$

$$\partial_{xx}\psi_2|_{z=0} = (-ik_x)^2 B_2 e^{i(\omega t - k_x x)} = -k_x^2 B_2 e^{i(\omega t - k_x x)} \quad (81)$$

$$\partial_{zz}\psi_2|_{z=0} = (ik_{z\beta_2})^2 B_2 e^{i(\omega t - k_x x)} = -k_{z\beta_2}^2 B_2 e^{i(\omega t - k_x x)} \quad (82)$$

Substitute the derivatives and cancel the common terms, this yields:

$$\boxed{\mu_1[2k_x k_{z\alpha_1}(A_1 - A') + (k_{z\beta_1}^2 - k_x^2)B_1] = \mu_2[2k_x k_{z\alpha_2}A_2 + (k_{z\beta_2}^2 - k_x^2)B_2]} \quad (83)$$

$$-2\mu_1 k_x k_{z\alpha_1} A' + \mu_1(k_{z\beta_1}^2 - k_x^2)B_1 - 2\mu_2 k_x k_{z\alpha_2} A_2 - \mu_2(k_{z\beta_2}^2 - k_x^2)B_2 = -2\mu_1 k_x k_{z\alpha_1} A_1$$

Multiply both sides by -1:

$$\boxed{2\mu_1 k_x k_{z\alpha_1} A' + \mu_1(k_x^2 - k_{z\beta_1}^2)B_1 + 2\mu_2 k_x k_{z\alpha_2} A_2 - \mu_2(k_x^2 - k_{z\beta_2}^2)B_2 = 2\mu_1 k_x k_{z\alpha_1} A_1}$$

Boundary Condition 4: Normal Stress

$$\sigma_{zz1}|_{z=0} = \sigma_{zz2}|_{z=0} \quad (84)$$

$$\begin{aligned} \lambda_1(\partial_{xx}\phi_1 + \partial_{zz}\phi_1) + 2\mu_1(\partial_{zz}\phi_1 + \partial_{xz}\psi_1)|_{z=0} \\ = \lambda_2(\partial_{xx}\phi_2 + \partial_{zz}\phi_2) + 2\mu_2(\partial_{zz}\phi_2 + \partial_{xz}\psi_2)|_{z=0} \end{aligned} \quad (85)$$

Computing:

$$\partial_{xx}\phi_1|_{z=0} = -k_x^2(A_1 + A')e^{i(\omega t - k_x x)} \quad (86)$$

$$\partial_{zz}\phi_1|_{z=0} = -k_{z\alpha_1}^2(A_1 + A')e^{i(\omega t - k_x x)} \quad (87)$$

$$\partial_{xz}\psi_1|_{z=0} = -k_x k_{z\beta_1} B_1 e^{i(\omega t - k_x x)} \quad (88)$$

$$\partial_{xx}\phi_2|_{z=0} = -k_x^2 A_2 e^{i(\omega t - k_x x)} \quad (89)$$

$$\partial_{zz}\phi_2|_{z=0} = -k_{z\alpha_2}^2 A_2 e^{i(\omega t - k_x x)} \quad (90)$$

$$\partial_{xz}\psi_2|_{z=0} = k_x k_{z\beta_2} B_2 e^{i(\omega t - k_x x)} \quad (91)$$

After substitution and simplification:

$$\begin{aligned} \lambda_1(-k_x^2 - k_{z\alpha_1}^2)(A_1 + A') + 2\mu_1(-k_{z\alpha_1}^2)(A_1 + A') - 2\mu_1 k_x k_{z\beta_1} B_1 \\ = \lambda_2(-k_x^2 - k_{z\alpha_2}^2)A_2 + 2\mu_2(-k_{z\alpha_2}^2)A_2 + 2\mu_2 k_x k_{z\beta_2} B_2 \end{aligned} \quad (92)$$

This can be rewritten as:

$$\begin{aligned} [\lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2](A_1 + A') + 2\mu_1 k_x k_{z\beta_1} B_1 \\ = [\lambda_2(k_x^2 + k_{z\alpha_2}^2) + 2\mu_2 k_{z\alpha_2}^2]A_2 - 2\mu_2 k_x k_{z\beta_2} B_2 \end{aligned} \quad (93)$$

$$\begin{aligned} \left[\lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2 \right] A' + 2\mu_1 k_x k_{z\beta_1} B_1 - \left[\lambda_2(k_x^2 + k_{z\alpha_2}^2) + 2\mu_2 k_{z\alpha_2}^2 \right] A_2 + 2\mu_2 k_x k_{z\beta_2} B_2 \\ = - \left[\lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2 \right] A_1 \end{aligned} \quad (94)$$

Let:

$$M_1 = \lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2$$

$$M_2 = \lambda_2(k_x^2 + k_{z\alpha_2}^2) + 2\mu_2 k_{z\alpha_2}^2$$

$$M_1 A' + 2\mu_1 k_x k_{z\beta_1} B_1 - M_2 A_2 + 2\mu_2 k_x k_{z\beta_2} B_2 = -M_1 A_1$$

Matrix Form (Zoeppritz System)

Collecting the four equations from the boundary conditions into matrix form, $\mathbf{M}x = \mathbf{b}$, where $\mathbf{b}$ is the right-hand vector containing the incident-wave terms (proportional to A_1). Explicitly the coefficient matrix (rows are eqns from the BCs, columns unknowns $[A', B_1, A_2, B_2]$) is:

$$\begin{pmatrix} k_x & -k_{z\beta 1} & -k_x & -k_{z\beta 2} \\ -k_{z\alpha 1} & -k_x & -k_{z\alpha 2} & k_x \\ 2\mu_1 k_x k_{z\alpha 1} & \mu_1(k_x^2 - k_{z\beta 1}^2) & 2\mu_2 k_x k_{z\alpha 2} & -\mu_2(k_x^2 - k_{z\beta 2}^2) \\ M_1 & 2\mu_1 k_x k_{z\beta 1} & -M_2 & 2\mu_2 k_x k_{z\beta 2} \end{pmatrix} \begin{pmatrix} A' \\ B_1 \\ A_2 \\ B_2 \end{pmatrix} = \begin{pmatrix} -k_x A_1 \\ -k_{z\alpha 1} A_1 \\ 2\mu_1 k_x k_{z\alpha 1} A_1 \\ -M_1 A_1 \end{pmatrix}$$

Divide the entire system by A_1 :

$$\begin{pmatrix} k_x & -k_{z\beta 1} & -k_x & -k_{z\beta 2} \\ -k_{z\alpha 1} & -k_x & -k_{z\alpha 2} & k_x \\ 2\mu_1 k_x k_{z\alpha 1} & \mu_1(k_x^2 - k_{z\beta 1}^2) & 2\mu_2 k_x k_{z\alpha 2} & -\mu_2(k_x^2 - k_{z\beta 2}^2) \\ M_1 & 2\mu_1 k_x k_{z\beta 1} & -M_2 & 2\mu_2 k_x k_{z\beta 2} \end{pmatrix} \begin{pmatrix} R_{PP} \\ R_{PS} \\ T_{PP} \\ T_{PS} \end{pmatrix} = \begin{pmatrix} -k_x \\ -k_{z\alpha 1} \\ 2\mu_1 k_x k_{z\alpha 1} \\ -M_1 \end{pmatrix}$$

Where:

$$M_1 = \lambda_1(k_x^2 + k_{z\alpha 1}^2) + 2\mu_1 k_{z\alpha 1}^2$$

$$M_2 = \lambda_2(k_x^2 + k_{z\alpha 2}^2) + 2\mu_2 k_{z\alpha 2}^2$$

The reflection and transmission coefficients are therefore:

$$R_{PP} = \frac{A'}{A_1}, \quad R_{PS} = \frac{B_1}{A_1}, \quad T_{PP} = \frac{A_2}{A_1}, \quad T_{PS} = \frac{B_2}{A_1}.$$

This 4×4 linear system is the general **Zoeppritz equation** for a P-wave incident on a plane elastic interface. Solving for RC and TC gives:

$$\begin{pmatrix} R_{PP} \\ R_{PS} \\ T_{PP} \\ T_{PS} \end{pmatrix} = \mathbf{M}^{-1} \mathbf{b}.$$

Where:

$$\mathbf{M} = \begin{pmatrix} k_x & -k_{z\beta 1} & -k_x & -k_{z\beta 2} \\ -k_{z\alpha 1} & -k_x & -k_{z\alpha 2} & k_x \\ 2\mu_1 k_x k_{z\alpha 1} & \mu_1(k_x^2 - k_{z\beta 1}^2) & 2\mu_2 k_x k_{z\alpha 2} & -\mu_2(k_x^2 - k_{z\beta 2}^2) \\ M_1 & 2\mu_1 k_x k_{z\beta 1} & -M_2 & 2\mu_2 k_x k_{z\beta 2} \end{pmatrix}$$

$$\mathbf{b} = \begin{pmatrix} -k_x \\ -k_{z\alpha 1} \\ 2\mu_1 k_x k_{z\alpha 1} \\ -M_1 \end{pmatrix}$$

The Zoeppritz system of equation above in its potential-based form can be transformed into a more compact form, similar to the one presented in the classical book by Aki and Richards

7. Special Case: P-Sv R/T Coefficients at Free Surface (Earth's Surface)

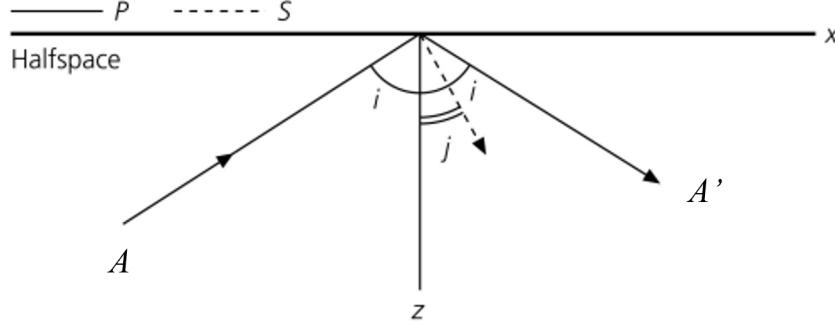


Figure 2: P-SV R/T Coefficients at Earth Surface

The free surface (e.g., the Earth's surface) is a special case where Medium 2 is a vacuum. This simplifies the problem because there are no transmitted waves and the boundary conditions become **stress-free**.

At Earth's surface, the boundary conditions simplify to **stress-free conditions**:

$$\sigma_{xz}|_{z=0} = 0, \quad \sigma_{zz}|_{z=0} = 0 \quad (95)$$

This gives us two equations for two unknowns (A' , B_1):

Free Surface Boundary Conditions

- Medium 1: Elastic solid with parameters $\alpha, \beta, \rho, \lambda, \mu$
- Medium 2: Vacuum ($z > 0$)
- Boundary conditions at $z = 0$: $\sigma_{zz} = 0$ and $\sigma_{xz} = 0$

Shear Stress = 0

From Boundary Condition 3 with medium 2 absent:

$$\mu[2k_x k_{z\alpha}(A_1 - A') + (k_{z\beta}^2 - k_x^2)B_1] \quad (96)$$

$$2k_x k_{z\alpha}(A_1 - A') + (k_x^2 - k_{z\beta}^2)B_1 = 0 \quad (97)$$

Dividing by k_x^2 :

$$2\frac{k_{z\alpha}}{k_x}(A_1 - A') + \left(\frac{k_{z\beta}^2}{k_x^2} - 1\right)B_1 = 0 \quad (98)$$

Defining:

$$\boxed{r_\alpha = \frac{k_{z\alpha}}{k_x} = \cot i_\alpha \quad \text{and} \quad r_\beta = \frac{k_{z\beta}}{k_x} = \cot i_\beta}$$

Note that $\frac{k_{z\beta}^2}{k_x^2} = r_\beta^2$, so:

$$\boxed{2r_\alpha(A_1 - A') + (r_\beta^2 - 1)B_1 = 0} \quad (99)$$

Normal Stress = 0

From Boundary Condition 4 with medium 2 absent:

$$\lambda(k_x^2 + k_{z\alpha}^2)(A_1 + A') + 2\mu(k_{z\alpha}^2)(A_1 + A') + 2\mu k_x k_{z\beta} B_1 = 0$$

$$[\lambda(k_x^2 + k_{z\alpha}^2) + 2\mu k_{z\alpha}^2](A_1 + A') + 2\mu k_x k_{z\beta} B_1 = 0 \quad (100)$$

Using the relations:

- $\lambda + 2\mu = \rho\alpha^2$

- $\mu = \rho\beta^2$
- $k_x^2 + k_{z\alpha}^2 = \omega^2/\alpha^2 = k_\alpha^2$

We can rewrite:

$$[\lambda(k_x^2 + k_{z\alpha}^2) + 2\mu k_{z\alpha}^2](A_1 + A') = 2\mu k_x k_{z\beta} B_1 \quad (101)$$

Dividing by k_x^2 :

$$\left[\lambda \left(1 + \frac{k_{z\alpha}^2}{k_x^2} \right) + 2\mu \frac{k_{z\alpha}^2}{k_x^2} \right] (A_1 + A') = 2\mu \frac{k_{z\beta}}{k_x} B_1 \quad (102)$$

$$[\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2](A_1 + A') = 2\mu r_\beta B_1 \quad (103)$$

This can be rewritten as:

$$\lambda(1 + r_\alpha^2)(A_1 + A') + 2\mu r_\alpha^2(A_1 + A') + 2\mu r_\beta B_1 = 0$$

$$\boxed{\lambda(1 + r_\alpha^2)(A_1 + A') + 2\mu (r_\alpha^2(A_1 + A') + r_\beta B_1) = 0} \quad (104)$$

Solving for Reflection Coefficients

System of Equations

From Equation 99:

$$2r_\alpha(A_1 - A') + (r_\beta^2 - 1)B_1 = 0$$

$$\boxed{A_1 - A' = -\frac{(r_\beta^2 - 1)}{2r_\alpha} B_1 = \frac{(1 - r_\beta^2)}{2r_\alpha} B_1} \quad (105)$$

From Equation 104:

$$\lambda(1 + r_\alpha^2)(A_1 + A') + 2\mu (r_\alpha^2(A_1 + A') + r_\beta B_1) = 0$$

$$[\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2] (A_1 + A') = -2\mu r_\beta B_1$$

$$\boxed{A_1 + A' = \frac{-2\mu r_\beta B_1}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2}} \quad (106)$$

Solving by Elimination Method

Let:

$$S = A_1 + A' \quad (\text{sum}) \quad (107)$$

$$D = A_1 - A' \quad (\text{difference}) \quad (108)$$

Then: $A_1 = (S + D)/2$ and $A' = (S - D)/2$

From Eq. 105:

$$D = \frac{(1 - r_\beta^2)}{2r_\alpha} B_1 \quad (109)$$

From Eq. 106:

$$S = -\frac{2\mu r_\beta}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2} B_1 \quad (110)$$

Therefore:

$$A_1 = \frac{1}{2} \left[\frac{(1 - r_\beta^2)}{2r_\alpha} - \frac{2\mu r_\beta}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2} \right] B_1 \quad (111)$$

$$A' = \frac{1}{2} \left[-\frac{2\mu r_\beta}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2} - \frac{(1 - r_\beta^2)}{2r_\alpha} \right] B_1 \quad (112)$$

Final Expressions

After significant algebraic manipulation, the reflection coefficients are:

P-to-P Reflection Coefficient:

$$R_P = \frac{A'}{A_1} = \frac{4r_\alpha r_\beta - (r_\beta^2 - 1)^2}{4r_\alpha r_\beta + (r_\beta^2 - 1)^2} \quad (113)$$

P-to-SV Conversion Coefficient:

$$R_{SV} = \frac{B_1}{A_1} = \frac{4r_\alpha(1 - r_\beta^2)}{4r_\alpha r_\beta + (r_\beta^2 - 1)^2} \quad (114)$$

where:

$$r_\alpha = \frac{k_{z\alpha}}{k_x} = \cot i_\alpha = \frac{\cos i_\alpha}{\sin i_\alpha} \quad (115)$$

$$r_\beta = \frac{k_{z\beta}}{k_x} = \cot i_\beta = \frac{\cos i_\beta}{\sin i_\beta} \quad (116)$$

Note: The angles i_α and i_β are measured from the vertical (normal to the surface).

8. Physical Interpretation

Key Features

1. **Mode Conversion:** Unlike SH waves, P-waves generate both P and SV reflected waves at a free surface.
2. **Critical Angles:** When $i > i_c = \sin^{-1}(\alpha/\beta)$, the SV wave becomes inhomogeneous (evanescent).
3. **Amplitude Behavior:**
 - At normal incidence ($i = 0$): $r_\alpha \rightarrow \infty$, $r_\beta \rightarrow \infty$, giving $R_P = -1$, $R_{SV} = 0$
 - At grazing incidence ($i \rightarrow 90$): $r_\alpha \rightarrow 0$, $r_\beta \rightarrow 0$, giving $R_P \rightarrow 1$, $R_{SV} \rightarrow 0$
 - Maximum P-to-SV conversion occurs at intermediate angles
4. **Energy Conservation:**

$$|R_P|^2 + |R_{SV}|^2 \frac{I_\beta \cos j_\beta}{I_\alpha \cos i_\alpha} = 1 \quad (117)$$

where $I = \rho v$ is the impedance.

Special Values

At normal incidence ($i_\alpha = 0$), using L'Hôpital's rule or direct evaluation:

$$R_P(0) = -1 \quad (\text{phase reversal}) \quad (118)$$

$$R_{SV}(0) = 0 \quad (\text{no mode conversion}) \quad (119)$$

At the critical angle $i_c = \sin^{-1}(\alpha/\beta)$:

- The SV wave grazes along the surface ($i_\beta = 90$)
- $r_\beta = 0$
- Maximum mode conversion occurs

9. Shuey's Approximation (Linearized AVO)

For small contrasts across an interface and weak angles, the P-wave reflection coefficient can be approximated:

$$\boxed{R(\theta) = R(0) + G \sin^2 \theta + F(\tan^2 \theta - \sin^2 \theta)} \quad (120)$$

where:

Intercept (normal incidence reflectivity):

$$R(0) = \frac{1}{2} \left(\frac{\Delta\rho}{\rho} + \frac{\Delta\alpha}{\alpha} \right) \quad (121)$$

Gradient (AVO slope):

$$G = \frac{\Delta\alpha}{2\alpha} - \frac{2\beta^2}{\alpha} \left(\frac{\Delta\rho}{\rho} + \frac{2\Delta\beta}{\beta} \right) \quad (122)$$

Large-angle term:

$$F = \frac{\Delta\alpha}{2\alpha} \quad (123)$$

Here:

- $\Delta\rho = \rho_2 - \rho_1$, $\Delta\alpha = \alpha_2 - \alpha_1$, $\Delta\beta = \beta_2 - \beta_1$
- $\rho = (\rho_1 + \rho_2)/2$, $\alpha = (\alpha_1 + \alpha_2)/2$, $\beta = (\beta_1 + \beta_2)/2$
- θ is the angle of incidence

This approximation is widely used in seismic exploration for amplitude-versus-offset (AVO) analysis to detect hydrocarbons and characterize lithology.

Physical Meaning of AVO Parameters

- $R(0)$: Measures the impedance contrast at normal incidence
- G : Captures the angle-dependent behavior, sensitive to fluid content and Poisson's ratio changes
- F : Important at large angles, related to P-wave velocity contrast

Summary

The P-SV problem is significantly more complex than the SH problem due to:

1. Mode conversion between P and SV waves
2. Coupling through both displacement and stress boundary conditions
3. Four coupled equations instead of two
4. Angle-dependent behavior that depends on both α and β

The exact Zoeppritz equations provide the complete solution, while Shuey's approximation offers a practical, linearized alternative for seismic interpretation.